



Monthly electricity generation using solar PV modules (Source: Prof Arun Kumar, AHEC, IIT Roorkee)

the net energy generation crossed 7000MWh by July 2017. The net reduction in carbon dioxide emission of the campus was 7208 tonnes till July 2017.

The 440,000pd solar water heating system covers 12 hostels, two guest houses and 14 residential areas. The heaters are connected with grid-connected geysers for heating during poor sunlight conditions. Assuming a complete 150 days-a-year usage of the solar water system, a 100pd system can save 4kWh energy every day. IIT Roorkee saves 2.6 million units electricity per year with a net carbon dioxide reduction of 2132 tonnes.

The solar cooking setup comprises 61 dish-type parabolic concentrator plates installed across nine student messes, covering an area of 976sq.m. The setup cuts down the conventional 14.2kg LPG cylinders usage by 5000 units.

The final benefit. The complete setup has enabled the institute to slash its electricity bill by an average of over 15 per cent and carbon footprint by almost 234 tonnes per year.

Case 2: Delhi International Airport

The Delhi International Airport Limited (DIAL) has also made a significant transition towards solar setup for electricity savings and an overall carbon-clean environment.

In an effort to effectively utilise the large proportion of its land at the Indra Gandhi International Airport, which would otherwise go unutilised commercially, it decided to dedicate a part of the space to solar arrangements.

ROI and expansion plans. DIAL installed 2.14MW solar setup in 2014 across its 72,843sq.m (18-acre) land including solar PV installations near the runway area, which required special permissions. Precautions for glare and radiation effects on flyers were examined and adopted in the setup. According to news sources, the total investment in the project till now has been ₹ 470 million, while electricity expenses of DIAL are around ₹ 2300 million per annum.

Considering the level cost of electricity (LCOE) over grid as ₹ 9.25 and that of solar power as ₹ 6.89, DIAL is saving ₹ 2.36 per unit using the solar setup. As a result, it is saving up to 20 per cent of the electricity expenses. Looking at the trend of up to 7 per cent annual increase in the electricity price, the arrangement will yield ROI in around six years.

The project is expanding with installation of another 5.7MW solar system setup. The solar set up expansion is expected to save ₹ 120 million annually in electricity cost, which will be an additional 5 per cent saving over the current power expenses.

To sum up

Not just industrial players but academic institutions too can gain significantly from solar setup investment. Those who go green get guaranteed returns in the form of cost savings while also contributing to a sustainable future for the coming generations. **EFY**

—Paromik Chakraborty, technical journalist, EFY

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India is on a mission to electronically connect all its strata to build a smarter infrastructure. Many stalwarts as well as new players are collectively taking up the responsibility to drive this smart revolution with their solutions that not only save costs and resources but also present great business opportunities. Let's take a look at some of them.

Energy management

Among the many solutions that promise to cut energy consumption or manage energy better, we came across Energy Bot's Smart ePlug. Running on Wi-Fi or the Internet, this product enables users to monitor and control the energy consumption of devices. The eBots app lets users reconcile the energy consumed and energy bill to know where most of the energy is being spent. Smart ePlug costs ₹ 2000 to ₹ 3500.